

Recovery Exam Statistical Mechanics

29 January 2025

Welcome to the final exam of Statistical Mechanics of the GEMIF. The exam covers all the topics we have seen during the course.

Question 1: [1.0 points]

Suppose a public health emergency is declared due to a pandemic (e.g., COVID-19), and on the terrace of a bar, 80% of the customers wear a mask 90% of the time, while the remaining 20% wear a mask only 50% of the time.

- (a) The police are instructed to shut down the terrace service if an inspection finds more than 15% of the customers without their mask on. Would you expect the terrace service to be shut down in the event of a police inspection? Justify your answer.
- (b) What is the probability that an individual who is not wearing a mask belongs to the group of customers who wear the mask 90% of the time?

Solution Question 1

- (a) Let A denote the event "wearing the mask 90% of the time," B denote the event "wearing the mask 50% of the time," $+$ denote the event "wearing the mask at the time of the police inspection," and $-$ denote the complementary event to $+$ (not wearing the mask).

From the given data:

$$P(A) = 0.8, \quad P(B) = 0.2,$$

$$P(+|A) = 0.9, \quad P(-|A) = 0.1, \quad P(+|B) = 0.5, \quad P(-|B) = 0.5.$$

The probability $P(+)$ that an individual on the terrace is wearing a mask at the time of the inspection is:

$$P(+) = P(+|A)P(A) + P(+|B)P(B) = (0.9)(0.8) + (0.5)(0.2) = 0.82 < 0.85.$$

so it is likely that the terrace service would be shut down during a police inspection.

- (b) For the second question, we are asked to compute $P(A|-)$. Using Bayes' theorem:

$$P(A|-) = \frac{P(-|A)P(A)}{P(-)}.$$

From the data:

$$P(-|A) = 0.1, \quad P(A) = 0.8, \quad P(-) = 0.18.$$

Thus:

$$P(A|-) = \frac{(0.1)(0.8)}{0.18} = \frac{0.08}{0.18} = 0.44.$$

This means that there is a 44% probability that an individual not wearing a mask belongs to the group of customers who wear their mask 90% of the time.

Question 2:

Suppose a classical ideal gas in three dimensions in the grand canonical ensemble (i.e. the system is in equilibrium with a reservoir at temperature T and chemical potential μ). Using the Gibbs definition of entropy, express S in terms of the chemical potential μ , k_B , the temperature of the system T and the expected number of gas particles $\langle N \rangle$. What is the maximum value that the chemical potential can have as a function of T ?

Solution Question 2:

(see attached pdf file)

Question 3

Consider a closed population of N individuals, where each individual can be in one of two states: Susceptible (S) the individual can contract the disease; Infected (I) the individual has the disease and can infect others. The system evolves with the following transition rules: A susceptible individual transitions to the infected state with a rate β , which depends on the fraction of infected individuals, I/N . An infected individual recovers and transitions back to the susceptible state with a constant rate γ .

Let n represent the number of infected individuals at time t , and let $p(n, t)$ be the probability that there are n infected individuals at time t .

- (a) Derive the master equation governing the time evolution of $p(n, t)$.

- (b) Specify the boundary conditions for n , i.e. ($n = 0$, $n = N$).
- (c) Prove that the stationary distribution

$$p_{\text{st}}(n) = C \prod_{i=1}^n \frac{\beta \frac{1}{N} (N - i + 1)}{\gamma i},$$

where C is a normalization constant, satisfies the stationary Master Equation. Hint: Use proof by induction.

Solution Question 3:

- (a) Derive the master equation governing the time evolution of $p(n, t)$. Define the possible transitions as:

- $n \rightarrow n + 1$: A susceptible individual becomes infected. The rate is

$$T(n \rightarrow n + 1) = \beta \frac{n}{N} (N - n),$$

where $N - n$ is the number of susceptible individuals.

- $n \rightarrow n - 1$: An infected individual recovers. The rate is

$$T(n \rightarrow n - 1) = \gamma n.$$

The master equation for $p(n, t)$ is given by:

$$\frac{\partial p(n, t)}{\partial t} = T(n-1 \rightarrow n)p(n-1, t) + T(n+1 \rightarrow n)p(n+1, t) - [T(n \rightarrow n+1) + T(n \rightarrow n-1)]p(n, t).$$

Substituting the transition rates:

$$\frac{\partial p(n, t)}{\partial t} = \beta \frac{n-1}{N} (N - n + 1)p(n-1, t) + \gamma(n+1)p(n+1, t) - [\beta \frac{n}{N} (N - n) + \gamma n]p(n, t).$$

- (b) Specify the boundary conditions for n .

- At $n = 0$: The probability flux into $n = 0$ must equal the flux out. Since $n - 1$ is not valid, we set:

$$\frac{\partial p(0, t)}{\partial t} = \gamma p(1, t) - \beta \frac{0}{N} (N - 0)p(0, t).$$

Simplifying:

$$\frac{\partial p(0, t)}{\partial t} = \gamma p(1, t).$$

- At $n = N$: Similarly, for the upper boundary:

$$\frac{\partial p(N, t)}{\partial t} = \beta \frac{N-1}{N} (N - (N - 1))p(N - 1, t) - \gamma N p(N, t).$$

(c) Prove that the stationary distribution

$$p_{\text{st}}(n) = C \prod_{i=1}^n \frac{\beta \frac{i-1}{N} (N-i+1)}{\gamma i},$$

where C is a normalization constant, satisfies the stationary Master Equation. Hint: Use proof by induction.

At the stationary state, $\frac{\partial p(n,t)}{\partial t} = 0$, so the master equation becomes:

$$0 = \beta \frac{n-1}{N} (N-n+1) p_{\text{st}}(n-1) + \gamma (n+1) p_{\text{st}}(n+1) - \left[\beta \frac{n}{N} (N-n) + \gamma n \right] p_{\text{st}}(n).$$

Substitute the proposed stationary solution:

$$p_{\text{st}}(n) = C \prod_{i=1}^n \frac{\beta \frac{i-1}{N} (N-i+1)}{\gamma i}.$$

We can prove the solution by induction.

Step 1: Base Case ($n = 0$)

At $n = 0$, $p_{\text{st}}(0) = C$, and there is no flux from $n = -1$ (not defined). The master equation becomes:

$$0 = \gamma p_{\text{st}}(1) - \beta \frac{0}{N} (N-0) p_{\text{st}}(0),$$

which simplifies to:

$$p_{\text{st}}(1) = \frac{\beta}{\gamma N} (N) p_{\text{st}}(0) = \frac{\beta}{\gamma} p_{\text{st}}(0).$$

This matches the proposed solution for $n = 1$.

Step 2: Inductive Hypothesis ($n > 0$)

Assume that the relation is true for n , and express the product by explicitly the $n + 1$ term:

$$p_{\text{st}}(n+1) = p_{\text{st}}(n) \cdot \frac{\beta(N-n)}{\gamma(n+1)N}.$$

Substitute $p_{\text{st}}(n+1)$ into the second term of the master equation:

$$\gamma(n+1)p_{\text{st}}(n+1) = \gamma(n+1) \cdot \left(p_{\text{st}}(n) \cdot \frac{\beta(N-n)}{\gamma(n+1)N} \right).$$

Simplify:

$$\gamma(n+1)p_{\text{st}}(n+1) = \beta \frac{1}{N} (N-n) p_{\text{st}}(n).$$

Substitute $p_{\text{st}}(n-1)$

We do the same as before:

$$p_{\text{st}}(n-1) = p_{\text{st}}(n) \cdot \frac{\gamma n}{\beta(N-n+1)} N.$$

Substitute $p_{\text{st}}(n-1)$ into the first term of the master equation:

$$\beta \frac{n-1}{N} (N-n+1) p_{\text{st}}(n-1) = \beta \frac{n-1}{N} (N-n+1) \cdot \left(p_{\text{st}}(n) \cdot \frac{\gamma n}{\beta \frac{n-1}{N} (N-n+1)} \right).$$

Simplify:

$$\beta \frac{n-1}{N} (N-n+1) p_{\text{st}}(n-1) = \gamma n p_{\text{st}}(n).$$

Substitute into the Master Equation.

Substitute all terms into the stationary master equation:

$$\beta \frac{n}{N} (N-n+1) p_{\text{st}}(n) + \gamma (n+1) p_{\text{st}}(n+1) - \left(\beta \frac{n+1}{N} (N-n-1) + \gamma (n+1) \right) p_{\text{st}}(n+1) = 0.$$

Then, we conclude:

$$0 = \frac{\beta p_{\text{st}}(n)}{N} \times (N-n) \times (1-n).$$

which is not generally true, and then this recurrence is not a solution of the Master Equation.

Problem 1 [2 points]

Consider a lattice with N sites. n_1 sites are occupied by one particle and have energy ϵ ; n_2 sites are occupied by two particles and have energy 2ϵ . Using the microcanonical ensemble:

- Calculate the entropy of the system in the thermodynamic limit in terms of the fraction of sites occupied with one particle x_1 .
- Obtain an expression for x_1 as a function of T . Can x_1 take any value between 0 and 1 at equilibrium? Explain why/why not?

Solution Problem 1:

(see pdf file)

Problem 2: [2 points]

Consider a perfect 2-dimensional crystal of surface S . Each of the N atoms of the crystal has a 'free electron' (spin $s = 1/2$) of mass m_e . These electrons behave as an ideal gas of non-interacting fermions.

- (a) Find the density of states of the gas of electrons $g(\epsilon)$.
- (b) Find the chemical potential as a function of temperature and find the Fermi energy.

Solution Problem 2:

(see pdf file)

Problem 3: [2.5 points]

Find the partition function of a classical ideal gas consisting of N point particles of mass m , in the presence of a gravitational field g , confined within a box of horizontal cross-section A and height $L = y_L - y_0$.

- (a) Calculate the internal energy
- (b) Calculate the heat capacity
- (c) Find the limits of the heat capacity for low and high values of $x = \beta mgL$, discussing their interpretation.

Solution Problem 3

- (a) In the Maxwell-Boltzmann approximation, the canonical partition function is expressed as:

$$Z = \frac{z^N}{N!},$$

where z is the canonical partition function of a single particle. The energy of a particle has two contributions: kinetic energy, which depends on the components of the momentum, and potential energy, which depends only on the y -component of the particle's position:

$$z = \frac{1}{h^3} \int d^3\vec{r} \int d^3\vec{p} \exp \left(-\beta \left(\frac{\vec{p}^2}{2m} + mgy \right) \right).$$

The integral over the momentum is

$$\frac{1}{h^3} \int d^3\vec{p} \exp \left(-\beta \frac{\vec{p}^2}{2m} \right) = \left(\frac{2m\pi}{\beta\hbar^2} \right)^{3/2}.$$

The presence of the gravitational field modifies the spatial integral, which is no longer simply the volume of the box:

$$\int d^3\vec{r} e^{-\beta mgy} = A \int_{y_0}^{y_L} dy e^{-\beta mgy} = A \frac{e^{-\beta mgy_0} - e^{-\beta mgy_L}}{\beta mg}.$$

Thus, the single-particle partition function becomes:

$$z = \left(\frac{2m\pi}{\beta\hbar^2} \right)^{3/2} \frac{A}{\beta mg} (e^{-\beta mgy_0} - e^{-\beta mgy_L}).$$

To calculate the internal energy $U = Nu$, we first compute $-\ln z$:

$$-\ln z = -\frac{3}{2} \ln\left(\frac{2m\pi}{\hbar^2}\right) + \frac{3}{2} \ln \beta - \ln A + \ln(\beta mg) - \ln(e^{-\beta mgy_0} - e^{-\beta mgy_L}).$$

From $u = -\frac{\partial \ln z}{\partial \beta}$, we obtain:

$$u = \frac{3}{2} k_B T + k_B T - \frac{-mgy_0 e^{-\beta mgy_0} + mgy_L e^{-\beta mgy_L}}{e^{-\beta mgy_0} - e^{-\beta mgy_L}}.$$

Using $y_L = y_0 + L$ and simplifying, we find:

$$u = \frac{3}{2} k_B T + k_B T + mgy_0 - \frac{mgL}{(e^{\beta mgL} - 1)}.$$

The first term, $\frac{3}{2} k_B T$, represents the average kinetic energy of the particle. The remaining terms represent the average potential energy.

(b) Now, the heat capacity $c_V = \frac{du}{dT}$ is:

$$c_V = \frac{3}{2} k_B + k_B \left(1 - \frac{x^2}{4 \sinh^2(x/2)} \right),$$

where $x = \beta mgL$ is the ratio of the potential energy difference between the top and bottom walls to the thermal energy $k_B T$.

(c) In the limits:

- For $x \ll 1$, $e^x - 1 \approx x + \frac{1}{2}x^2$, so $f(x) \approx \frac{1}{2}$, and $c_V \approx \frac{3}{2} k_B$.
- For $x \gg 1$, $e^x \gg 1$, $f(x) \approx \frac{1}{x}$, and $c_V \approx \frac{3}{2} k_B + k_B$.